STRhub Verified

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Technical Reproducibility Validation Report

GangSTR

Version 2.5
Autosomal STR

Verification date	2026-06-24
Repository commit	v2.5
Attestation level	**Installs**
Permalink	https://strhub.app/verified/gangstr-2-5
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution — confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**2/5 gates passed — Installs**
ATTESTATION LEVEL	**Installs**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	GangSTR
Version	2.5
Repository	https://github.com/gymreklab/gangstr
Commit (pinned)	v2.5
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-06-24 21:37:35 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/28131191634

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the BAM, reference, and BED paths with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · GangSTR 2.5 · commit v2.5
$ GangSTR \
--bam \
/data/in/input.bam \
--ref \
/data/ref/hg38.fa \
--regions \
/data/in/regions.bed \
--out \
/data/out/output
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS

Runs	Executes end-to-end without crashing	—
Expected IO	Produces a non-empty file in the declared format	—
Output Structure Validation	Output structure matches expected genotype-bearing format	—
Execution log	gangstr-2-5.log-external.txt	

5. Out of Scope

This report does not evaluate any of the following:
– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	result.vcf.gz
Format	VCF
Sequence records	—
Distinct STR loci	—
Total reads	—
Max depth (single locus)	—

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below.

Dataset	Illumina BAM (hg38) — NA12878 (autosomal, female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	—

8. Verification Matrix

PASS	Reference dataset	Illumina BAM (hg38) — NA12878 (autosomal, female)
	README check (5/5)	Advisory — all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ **Runs end-to-end**

GangSTR 2.5 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 2 verification gates were passed.

◆ **Produces valid output**

The tool generated a structurally valid VCF file with genotype calls across 0 target forensic STR loci, with a maximum read depth of 0 at —.

◆ **No bundled demo data**

GangSTR does not include its own test or demo dataset. STRhub Verified supplied the GIAB NA12878 300× dataset (GRCh38) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.